

Table 1: Friedman ranks of the 48 GA component configurations over the 12 benchmark instances. Lower ranks indicate better performance.

Configuration	C1	C2	C3	C4	C5	C11	C12	G1	G5	G9	G13	G17
st-pmx-insm-bt	12	22	15	19	21	21	19	23	16	26	25	19
st-pmx-insm-rw	10	23	20	18	17	19	22	20	21	17	24	24
st-pmx-swpm-bt	24	17	17	22	26	24	24	17	17	22	18	22
st-pmx-swpm-rw	19	20	21	25	18	9	18	21	23	16	26	25
st-pmx-nom-bt	14	21	22	26	28	26	21	18	24	14	13	23
st-pmx-nom-rw	27	26	25	21	24	22	25	22	20	20	19	28
st-cx-insm-bt	25	29	32	33	36	30	29	32	32	34	32	34
st-cx-insm-rw	26	32	31	31	30	25	33	29	28	32	33	33
st-cx-swpm-bt	32	33	36	34	33	31	35	31	33	29	34	32
st-cx-swpm-rw	28	30	29	28	31	29	30	27	27	28	30	30
st-cx-nom-bt	44	39	33	35	35	38	37	34	31	36	36	35
st-cx-nom-rw	43	44	30	36	34	40	38	36	36	35	35	36
st-ox-insm-bt	11	16	9	15	12	11	15	9	10	15	14	14
st-ox-insm-rw	17	14	16	10	16	18	20	19	12	18	20	21
st-ox-swpm-bt	8	11	11	16	15	10	16	12	6	13	17	17
st-ox-swpm-rw	20	18	12	20	20	20	17	24	15	24	21	16
st-ox-nom-bt	13	19	13	12	13	15	13	13	8	21	12	13
st-ox-nom-rw	18	15	14	13	19	17	14	8	13	23	15	15
st-nox-insm-bt	36	36	38	38	37	39	36	40	40	40	39	39
st-nox-insm-rw	40	37	40	41	41	44	40	41	39	41	40	41
st-nox-swpm-bt	38	40	42	43	43	41	41	43	42	43	41	43
st-nox-swpm-rw	42	41	41	40	44	37	43	42	38	38	42	37
st-nox-nom-bt	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5
st-nox-nom-rw	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5
bs-pmx-insm-bt	15	13	7	11	9	14	6	10	11	7	7	9
bs-pmx-insm-rw	16	10	8	9	8	13	8	14	18	12	10	8
bs-pmx-swpm-bt	9	8	19	8	7	7	12	11	7	4	9	11
bs-pmx-swpm-rw	23	9	10	17	11	12	11	15	22	10	16	7
bs-pmx-nom-bt	22	7	18	7	10	16	9	16	14	9	8	10
bs-pmx-nom-rw	7	12	23	14	14	8	10	7	19	11	11	12
bs-cx-insm-bt	30	27	28	30	23	28	28	25	29	30	27	18
bs-cx-insm-rw	21	25	26	27	29	27	27	28	26	25	22	26
bs-cx-swpm-bt	29	28	27	24	27	32	26	30	30	31	28	27
bs-cx-swpm-rw	31	24	24	23	25	23	23	26	25	19	23	20
bs-cx-nom-bt	37	38	35	32	32	35	31	33	34	27	31	31
bs-cx-nom-rw	34	31	34	29	22	33	32	35	35	33	29	29
bs-ox-insm-bt	1	2	2	4	1	3	2	1	1	2	4	3
bs-ox-insm-rw	5	5	6	5	5	6	5	3	9	8	6	5
bs-ox-swpm-bt	3	3	3	2	2	1	1	5	2	1	1	1
bs-ox-swpm-rw	6	4	5	6	6	5	7	6	5	5	5	6
bs-ox-nom-bt	2	1	1	1	3	2	3	2	3	3	2	2
bs-ox-nom-rw	4	6	4	3	4	4	4	4	4	6	3	4
bs-nox-insm-bt	35	35	37	37	38	34	34	38	37	37	38	40
bs-nox-insm-rw	33	34	39	39	40	36	39	37	41	42	37	38
bs-nox-swpm-bt	41	42	43	44	42	43	42	39	44	44	44	42
bs-nox-swpm-rw	39	43	44	42	39	42	44	44	43	39	43	44
bs-nox-nom-bt	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5
bs-nox-nom-rw	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5

Friedman test summary. The omnibus Friedman test indicates a highly significant difference among the 48 GA configurations: $\chi_F^2(47) = 540.709$, $p = 4.105 \times 10^{-85}$, based on $N = 12$ benchmark instances and $k = 48$ configurations. Kendall's coefficient of concordance is $W = 0.959$, indicating very strong agreement in the relative ranking of the configurations across instances. The three lowest average ranks are obtained by **bs-ox-nom-bt** ($\bar{R} = 2.083$), **bs-ox-swpm-bt** ($\bar{R} = 2.083$), and **bs-ox-insm-bt** ($\bar{R} = 2.167$).